CCNA 200-301 Notes

Chapter 35 — Inter-VLAN Routing with a Layer 3 Switch

Author: Yousef Sorour

Based on the supplied video summary and Packet Tracer topology. This method uses Switch Virtual Interfaces (SVIs) on a multilayer switch instead of router sub-interfaces.

1. What Is Inter-VLAN Routing with a Layer 3 Switch?

A Layer 3 switch can perform both Layer 2 switching and Layer 3 routing. Each VLAN remains a separate Layer 2 broadcast domain, but an SVI can provide that VLAN's default gateway. With IP routing enabled, the switch routes packets between connected VLAN subnets.

Core workflow: Create VLANs → assign access ports → configure an SVI and gateway IP for each VLAN → enable ip routing → set each PC's default gateway.

2. Topology and Addressing

VLAN ID	Department	Network	SVI / Default Gateway
2	HR	192.168.1.0/24	192.168.1.1
3	IT	192.168.2.0/24	192.168.2.1
4	Accounting (Acc)	192.168.3.0/24	192.168.3.1

The screenshot labels VLANs 2, 3, and 4. The video summary also mentions VLAN 5, but it is not shown in the supplied topology, so the configuration below follows the visible three-VLAN layout.

3. Create and Name VLANs

Commands:

```
enable
configure terminal
vlan 2
name HR
exit
vlan 3
name IT
exit
vlan 4
name Accounting
exit
```

Purpose: create the Layer 2 VLANs and assign readable names. VLAN IDs determine membership; names are labels.

4. Assign Switch Ports to VLANs

The screenshot labels HR ports as 1–3, IT ports as 4–6, and Accounting ports as 7–9. Verify the mapping in Packet Tracer before applying these examples.

```
! HR - Fa0/1 through Fa0/3
interface range fastEthernet 0/1 - 3
switchport mode access
switchport access vlan 2
exit

! IT - Fa0/4 through Fa0/6
interface range fastEthernet 0/4 - 6
switchport mode access
switchport access vlan 3
exit

! Accounting - Fa0/7 through Fa0/9
interface range fastEthernet 0/7 - 9
switchport mode access
switchport access vlan 4
exit
```

Purpose of the port commands:

Command	Purpose
interface range fa0/1 - 3	Select multiple switch ports at once.
switchport mode access	Makes selected ports access ports for end devices.
switchport access vlan 2	Places selected ports in VLAN 2. Use the appropriate VLAN ID for each group.

5. Configure the SVIs (Default Gateways)

An SVI is a logical Layer 3 interface for a VLAN. Configure one SVI per VLAN that needs routing. Its IP address is normally the default gateway for devices in that VLAN.

```
interface vlan 2
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface vlan 3
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface vlan 4
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

Command purposes:

Command	Purpose
interface vlan 2	Enters or creates the SVI for VLAN 2.
ip address ...	Assigns the gateway IP and subnet mask for that VLAN.
no shutdown	Administratively enables the SVI.

6. Enable Layer 3 Routing — The Critical Command

```
configure terminal
ip routing
end
```

Purpose: enables IP routing between SVIs on the multilayer switch. Without this command on the platform/configuration shown in the lesson, the switch will not route traffic between VLAN subnets.

7. Example PC Addressing

Department	Example PC IP	Mask	Default Gateway
HR / VLAN 2	192.168.1.10	255.255.255.0	192.168.1.1
IT / VLAN 3	192.168.2.10	255.255.255.0	192.168.2.1
Accounting / VLAN 4	192.168.3.10	255.255.255.0	192.168.3.1

These host IPs are examples, not confirmed device settings. Each PC needs a unique IP in its VLAN subnet and the matching SVI as default gateway.

8. Verification Commands

Command	What to verify
show vlan brief	VLANs exist and ports belong to the intended VLANs.
show ip interface brief	SVI IP addresses and interface status.
show ip route	Connected routes for VLAN subnets appear after routing is configured.
show running-config	Review SVIs and routing configuration.
ping <gateway-IP>	Each PC should first ping its own VLAN gateway.

9. Troubleshooting Checklist

• Verify PC-port membership with show vlan brief.

• Verify each VLAN has an SVI with the correct IP/mask.

• Confirm ip routing is enabled in global configuration mode.

• Check unique PC IPs and correct default gateways.

• Ping the local gateway before testing cross-VLAN communication.

• If an SVI is down, check that the VLAN exists and an active Layer 2 port in that VLAN is up.

10. Exam Notes

• A Layer 3 switch performs switching and routing.
• An SVI (interface vlan X) provides the Layer 3 gateway for a VLAN.
• Each VLAN should use its own IP subnet in this design.
• ip routing enables routing between the directly connected VLAN interfaces.
• This topology does not need router sub-interfaces or a switch-to-router trunk when end devices connect directly to the multilayer switch.
• PC-facing ports should be access ports assigned to their correct VLAN.

11. Quick Revision

Create VLANs → assign access ports → configure interface vlan X + gateway IP → no shutdown → ip routing → set PC IP/mask/default gateway → verify and test cross-VLAN ping.

Scope note: The supplied video summary mentions VLAN 5, but the screenshot labels VLANs 2, 3, and 4 only. This document follows the screenshot. If VLAN 5 exists in your actual lab, create and assign it, configure its SVI with a separate subnet, and verify the addressing.

Reference: Cisco Catalyst 3560 documentation explains that SVIs provide a Layer 3 interface for VLANs and IP routing must be enabled for the switch to route between them.